SUBSTITUTION

BLAKE FARMAN

Lafayette College

Name: Solutions

Use substitution and the Fundamental Theorem of Calculus to evaluate the following integrals.

1. $\int_0^1 \cos(\pi t/2)\,\mathrm{d}t$

$u = \frac{\pi t}{2} \Rightarrow du = \frac{\pi}{2}dt$

$\Rightarrow \frac{2}{\pi}du = dt$

$$\int_0^1 \cos\left(\frac{\pi t}{2}\right)dt = \int_0^{\pi/2} \cos(u)\frac{2}{\pi}du$$

$$= \frac{2}{\pi}\int_0^{\pi/2}\cos(u)du$$

$$= \frac{2}{\pi}\sin(u)\Big|_0^{\pi/2}$$

$$= \frac{2}{\pi}\left[\sin(\pi/2) - \sin(0)\right]$$

$$= \frac{2}{\pi}(1) = \boxed{2/\pi}$$

2. $\int_0^1 (2t-1)^{50}\,\mathrm{d}t$

$u = 2t-1 \Rightarrow du = 2dt \Rightarrow \frac{1}{2}du = dt$

$$\int_0^1 (2t-1)^{50}dt = \int_{-1}^1 u^{50}\left(\frac{1}{2}\right)du$$

$$= \frac{1}{2}\int_{-1}^1 u^{50}du$$

$$= \frac{1}{2}\frac{u^{51}}{51}\Big|_{-1}^1$$

$$= \frac{1}{2}\left(\frac{1}{51}\right)(1-(-1)) = \frac{1}{2}\left(\frac{1}{51}\right)(2)$$

$$= \boxed{1/51}$$

3. $\int_0^{\pi/6} \frac{\sin(t)}{\cos^2(t)}\,\mathrm{d}t$

$$\int_0^{\pi/6}\frac{\sin(t)}{\cos^2(t)}dt = \int_0^{\pi/6}\sec(t)\tan(t)dt$$

$$= \sec(t)\Big|_0^{\pi/6}$$

$$= \sec(\pi/6) - \sec(0)$$

$$= \frac{1}{\cos(\pi/6)} - \frac{1}{\cos(0)}$$

$$= \boxed{\frac{2}{\sqrt{3}} - 1}$$

4. $\int_0^3 x\sqrt{9-x^2}\,\mathrm{d}x$

$u = 9-x^2 \Rightarrow du = -2xdx \Rightarrow -\frac{1}{2}du = xdx$

$$\int_0^3 x\sqrt{9-x^2}\,dx = \int_9^0 -\frac{1}{2}\sqrt{u}\,du$$

$$= -\frac{1}{2}\int_9^0 u^{1/2}du$$

$$= -\frac{1}{2}\left[\frac{2}{3}u^{3/2}\right]_9^0$$

$$= -\frac{1}{3}\left(0^{3/2} - 9^{3/2}\right)$$

$$= \frac{1}{3}\left(\sqrt{9}\right)^3$$

$$= \frac{1}{3}(3)^3$$

$$= 3^2$$

$$= \boxed{9}$$

5. $\displaystyle\int_0^{\pi/2} \cos(x)\sin(\sin(x))\,\mathrm{d}x$

$u = \sin(x) \Rightarrow du = \cos(x)\,dx$

$$\int_0^{\frac{\pi}{2}} \cos(x)\sin(\sin(x))\,dx = \int_0^1 \sin(u)\,du$$

$$= -\cos(u)\Big|_0^1$$

$$= -\cos(1) - (-\cos(0))$$

$$= \boxed{-\cos(1) + 1}$$

6. $\displaystyle\int_{-\pi/3}^{\pi/3} x^4 \sin(x)\,\mathrm{d}x$

$f(x) = x^4 \sin(x)$

$f(-x) = (-x)^4 \sin(-x) = x^4(-\sin(x)) = -x^4 \sin(x) = -f(x)$

$\Rightarrow x^4 \sin(x)$ is an odd function

$$\Rightarrow \int_{-\pi/3}^{\pi/3} x^4 \sin(x)\,dx = \boxed{0}$$